

Step - 1 : Procedure / Implementation

```
# Install required libraries
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn

Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,>=12.3.0) (1.3.0)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,>=12.3.0) (12.3.0)
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Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.24.3)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
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Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0) (2025.3.0)
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Requirement already satisfied: aiohttp!=4.0.0a0,!<4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0) (4.0.0a1)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0) (0.23.0)
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Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.12.0)
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Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
```

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer
from transformers import AutoModelForSequenceClassification
from transformers import Trainer, TrainingArguments
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

Step - 2 : Text Preprocessing

```
from datasets import load_dataset

# Load dataset
dataset = load_dataset("imdb")

# Create train and test variables
train_data = dataset["train"]
```

```
test_data = dataset["test"]
```

```
# Load pretrained tokenizer (DistilBERT)
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%                               483/483 [00:00<00:00, 9.86kB/s]

tokenizer_config.json: 100%                       48.0/48.0 [00:00<00:00, 677B/s]

vocab.txt:      232k/? [00:00<00:00, 2.73MB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
tokenizer.json:  466k/? [00:00<00:00, 4.35MB/s]
```

Step - 3 : Vocabulary Creation & Encoding

```
def tokenize_function(example):

    # Tokenize text
    return tokenizer(
        example["text"],
        padding="max_length",
        truncation=True,
        max_length=256
    )

# Apply tokenization to dataset
tokenized_train = train_data.map(tokenize_function, batched=True)
tokenized_test = test_data.map(tokenize_function, batched=True)
```

```
Map: 100%                               20000/20000 [00:31<00:00, 803.51 examples/s]
Map: 100%                               5000/5000 [00:05<00:00, 938.18 examples/s]
```

Step - 4 : Train-Test Split

```
dataset = load_dataset("imdb")
dataset = dataset["train"].train_test_split(test_size=0.2)
train_data = dataset["train"]
test_data = dataset["test"]
```

```
README.md:      7.81k/? [00:00<00:00, 163kB/s]

plain_text/train-00000-of-00001.parquet: 100%                21.0M/21.0M [00:01<00:00, 129MB/s]
plain_text/test-00000-of-00001.parquet: 100%                 20.5M/20.5M [00:01<00:00, 62.5MB/s]
plain_text/unsupervised-00000-of-00001.p(...): 100%          42.0M/42.0M [00:01<00:00, 43.0MB/s]

Generating train split: 100%                                25000/25000 [00:01<00:00, 24668.05 examples/s]
Generating test split: 100%                                25000/25000 [00:01<00:00, 22557.84 examples/s]
Generating unsupervised split: 100%                         50000/50000 [00:01<00:00, 45658.79 examples/s]
```

Step - 5 : Create PyTorch Dataset Format

```
tokenized_train.set_format(
    type="torch",
    columns=["input_ids", "attention_mask", "label"]
)

tokenized_test.set_format(
    type="torch",
    columns=["input_ids", "attention_mask", "label"]
)
```

Step - 6 : Load Transformer Model

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
```

```

    num_labels=2    # positive / negative
)

```

model.safetensors: 96%

256M/268M [00:06<00:00, 68.3MB/s]

Loading weights: 100%

100/100 [00:00<00:00, 321.67it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status	
-----+-----+		
vocab_transform.bias	UNEXPECTED	
vocab_transform.weight	UNEXPECTED	
vocab_layer_norm.bias	UNEXPECTED	
vocab_layer_norm.weight	UNEXPECTED	
vocab_projector.bias	UNEXPECTED	
classifier.weight	MISSING	
pre_classifier.weight	MISSING	
pre_classifier.bias	MISSING	
classifier.bias	MISSING	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task

Step - 7 : Define Evaluation Metrics

```

def compute_metrics(pred):

    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(
        labels,
        preds,
        average='binary'
    )

    acc = accuracy_score(labels, preds)

    return {
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }

```

step - 7 : Training Configuration

```

training_args = TrainingArguments(

    output_dir="./results",

    eval_strategy="epoch",

    learning_rate=2e-5,

    per_device_train_batch_size=8,

    per_device_eval_batch_size=8,

    num_train_epochs=2,

    weight_decay=0.01,

    logging_dir="./logs"
)

```

`logging\_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD\_LOGGING\_DIR` instead.

Step - 8 : Model Training

```

trainer = Trainer(

    model=model,

    args=training_args,

    train_dataset=tokenized_train,

    eval_dataset=tokenized_test,

```

```
        compute_metrics=compute_metrics
    )
```

```
# Train the model
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no pin_memory_device is provided. This warning can be disabled by setting pin_memory=False.
  super().__init__(loader)
```

 [144/5000 25:31 < 14:32:49, 0.09 it/s, Epoch 0.06/2]

Epoch	Training Loss	Validation Loss
-------	---------------	-----------------

Step - 9 : Model Evaluation

```
results = trainer.evaluate()
```

```
print(results)
```